Scaled DDM network sc_s1106_e245: SBC, six variability cases

EMC2 dev−nle 142bd497, N = 400 trials, 500 replicates per case, AWS run1 2026−10−15

Each case: a title page with the summary, then the rank−ECDF difference plots (95% envelope shaded)

case std bias (* = outside envelope)	**v**	**a**	**t0**	**Z**	**sv**	**SZ**	**st0**
2: st0 + SZ free (sv = 0)	−0.06	−0.03	+0.01	+0.08	−	−0.04	+0.05
3: st0 + sv free (SZ = 0)	+0.01	+0.02	+0.05	+0.05	+0.02	−	+0.02
4: sv + SZ free (st0 = 0)	−0.02	−0.04	+0.08	−0.05	+0.01	+0.04	−
5: st0 free (sv = SZ = 0)	+0.05	+0.05*	+0.05	−0.06	−	−	−0.00
7: SZ free (sv = st0 = 0)	+0.04	−0.03	+0.01	−0.04	−	−0.01	−
8: all free	−0.07	−0.10	+0.13	+0.01	+0.02	−0.00	+0.02

Coverage 0.932−0.968; min KS p 0.020; 1 of 35 parameter tests outside the envelope (a, case 5)

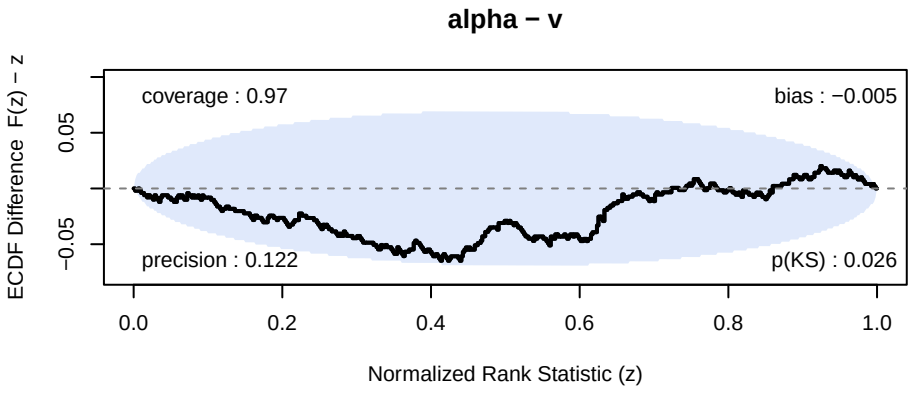
Old networks for comparison: case 2 a, st0; 3 a, st0; 4 a, t0; 5 t0, st0; 7 Z; 8 t0, Z (worst |std bias| .15−.50)

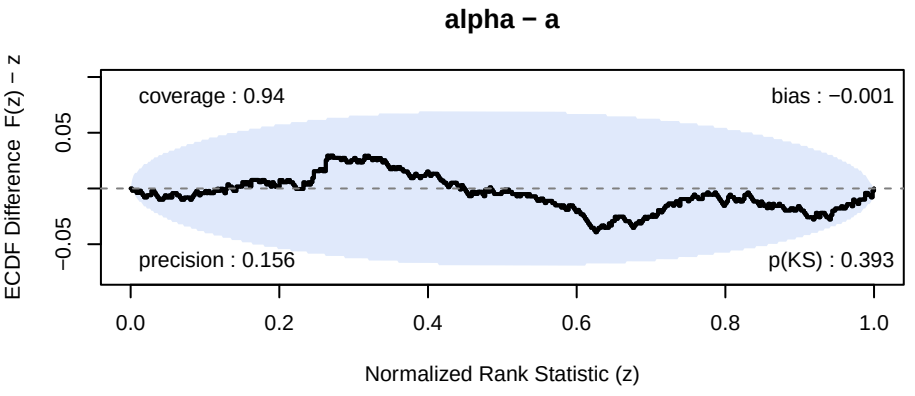
Case 2: st0 + SZ free (sv = 0)

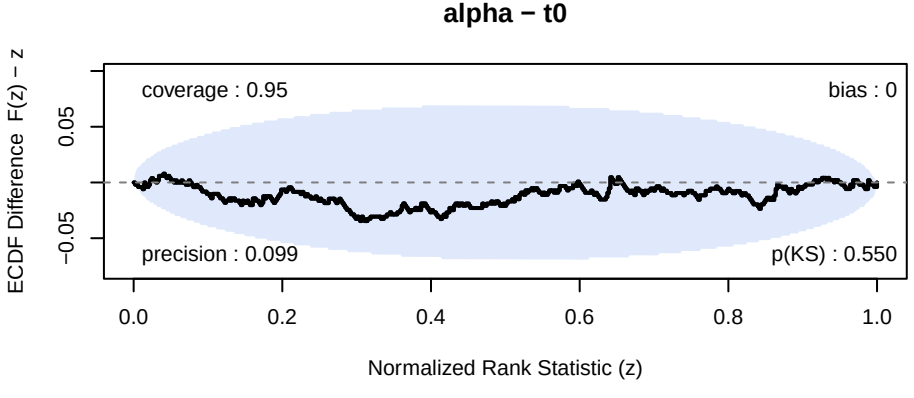
regime sv0

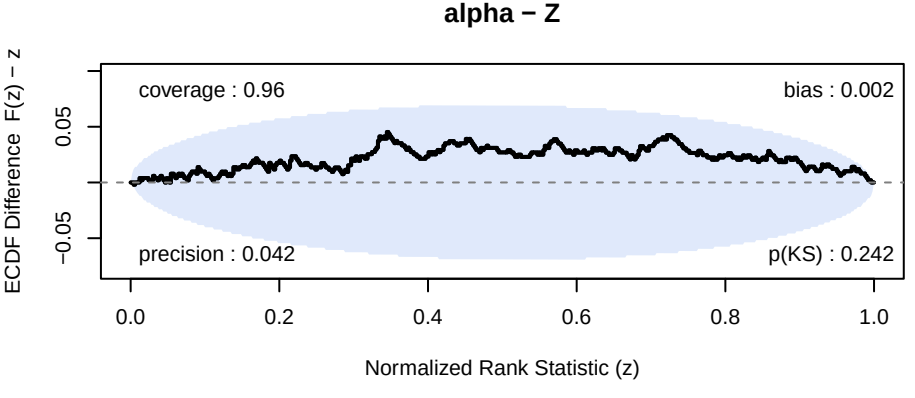
outside envelope: none

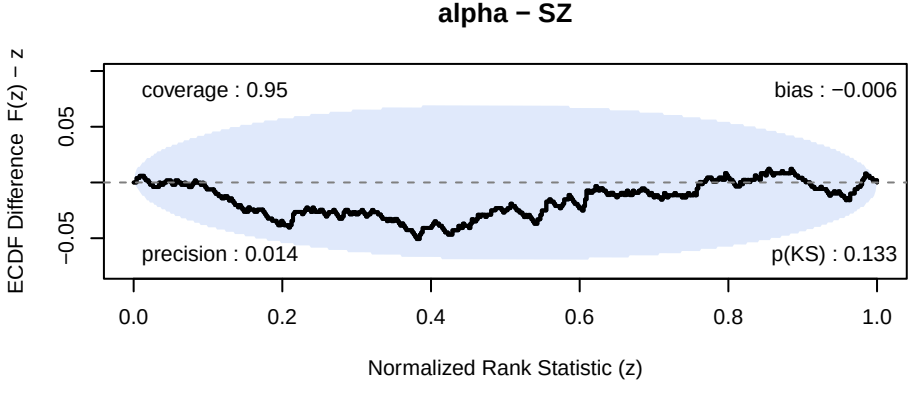
worst |std bias| 0.08 (Z); coverage 0.942−0.968; min KS p 0.026

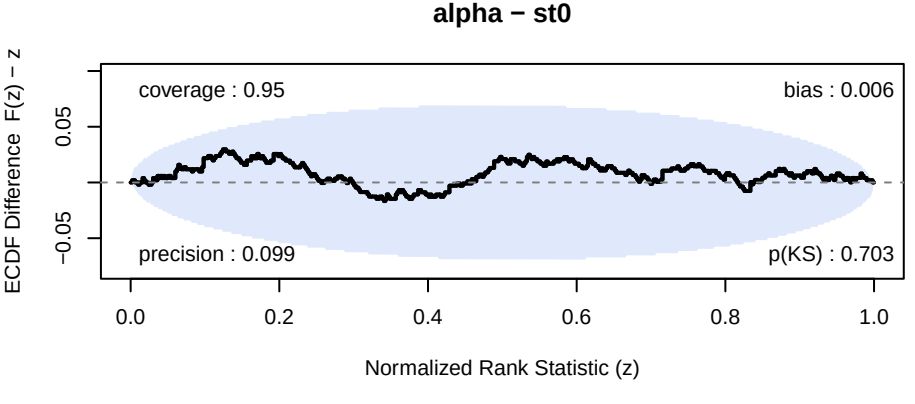

alpha − v
coverage : 0.97
bias : −0.005
precision : 0.122
p(KS) : 0.026
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − a
coverage : 0.94
bias : −0.001
precision : 0.156
p(KS) : 0.393
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.95
bias : 0
precision : 0.099
p(KS) : 0.550
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.96
bias : 0.002
precision : 0.042
p(KS) : 0.242
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − SZ
coverage : 0.95
bias : −0.006
precision : 0.014
p(KS) : 0.133
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

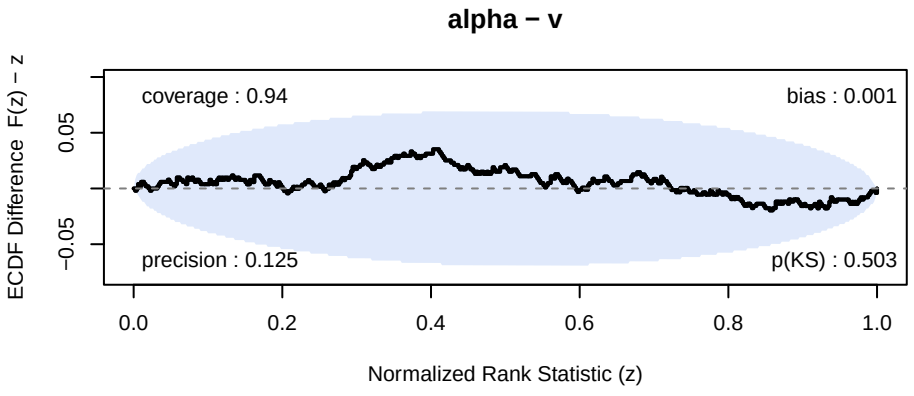

alpha − st0
coverage : 0.95
bias : 0.006
precision : 0.099
p(KS) : 0.703
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

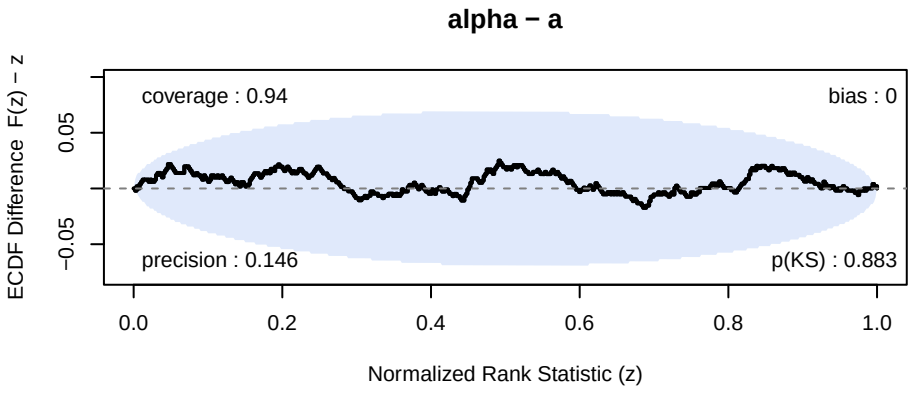
Case 3: st0 + sv free (SZ = 0)

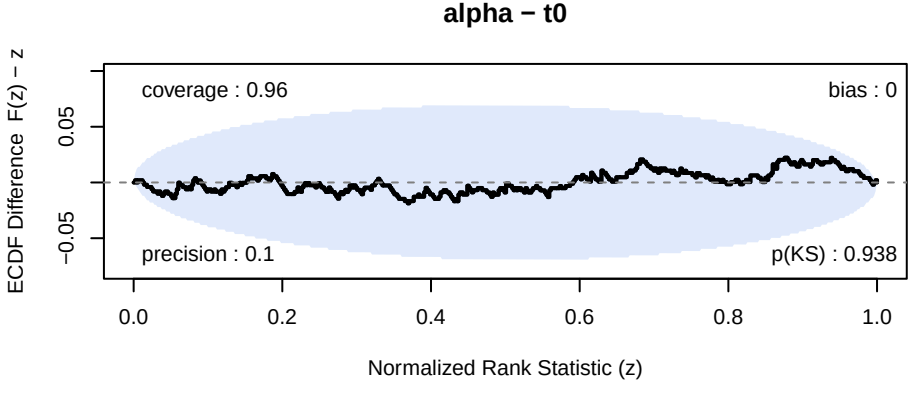
regime SZ0

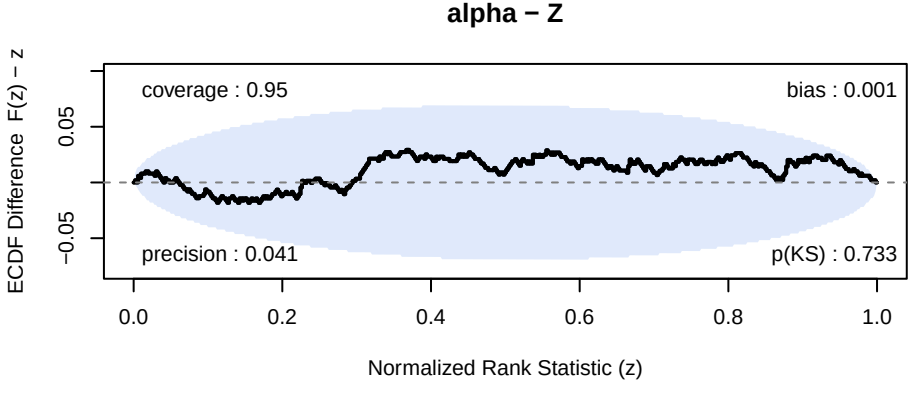
outside envelope: none

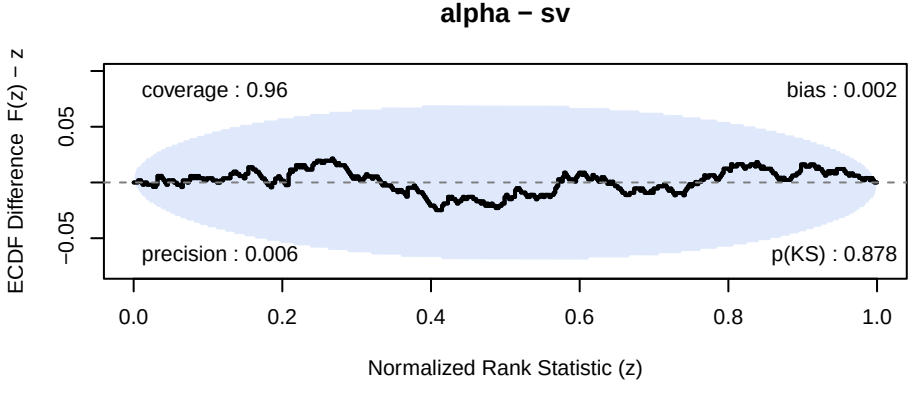
worst |std bias| 0.05 (t0); coverage 0.940−0.964; min KS p 0.503

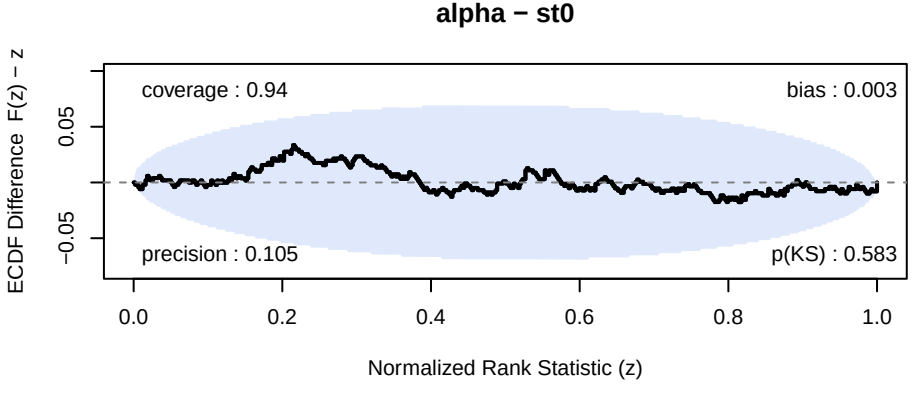

alpha − v
coverage : 0.94
bias : 0.001
precision : 0.125
p(KS) : 0.503
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − a
coverage : 0.94
bias : 0
precision : 0.146
p(KS) : 0.883
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.96
bias : 0
precision : 0.1
p(KS) : 0.938
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.95
bias : 0.001
precision : 0.041
p(KS) : 0.733
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − sv
coverage : 0.96
bias : 0.002
precision : 0.006
p(KS) : 0.878
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

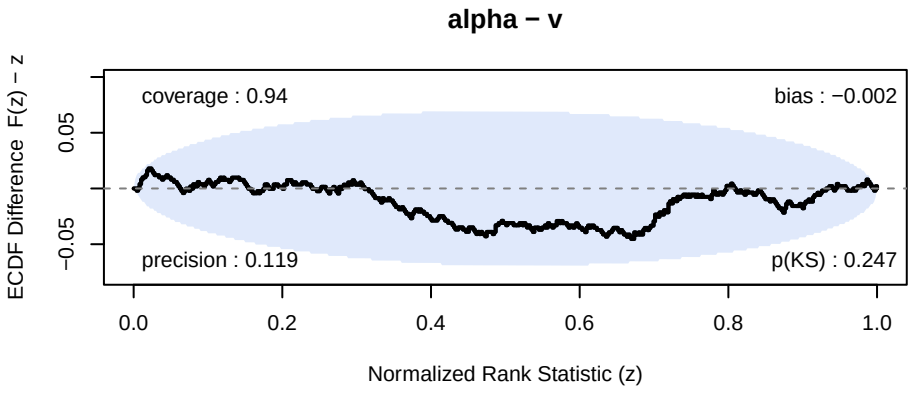

alpha − st0
coverage : 0.94
bias : 0.003
precision : 0.105
p(KS) : 0.583
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

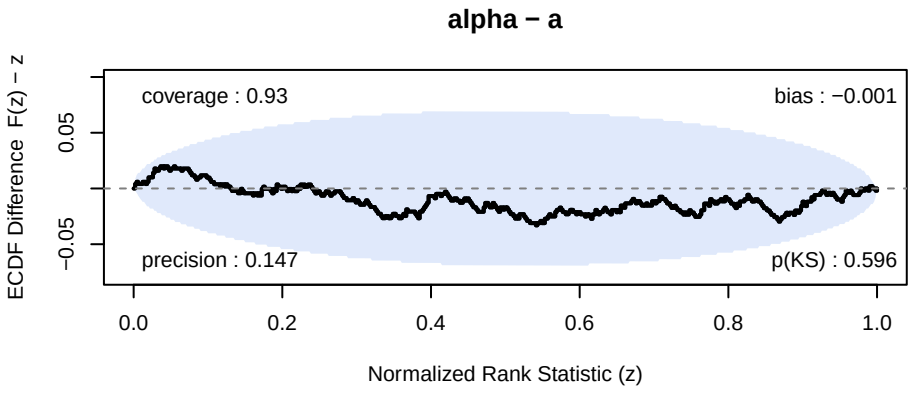
Case 4: sv + SZ free (st0 = 0)

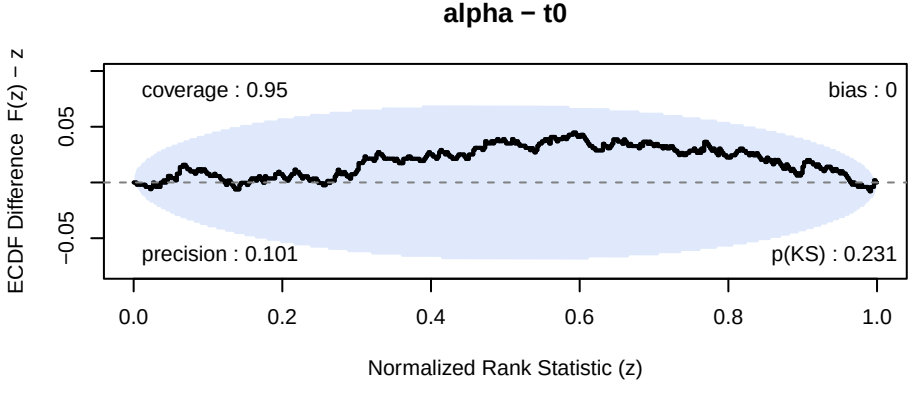
regime st00

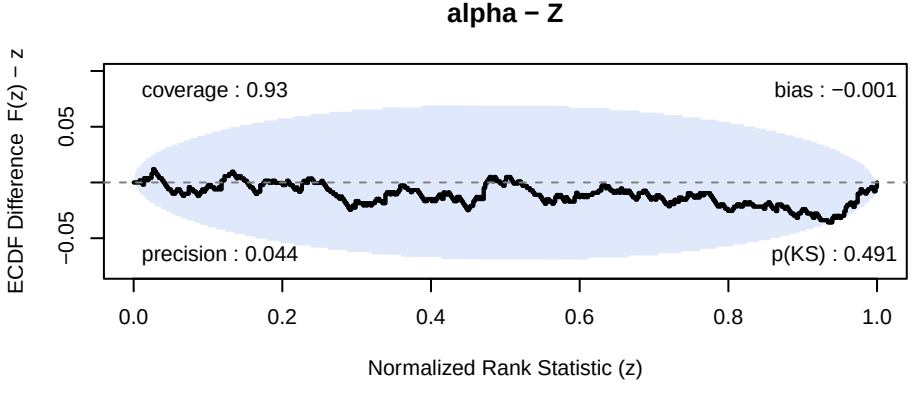
outside envelope: none

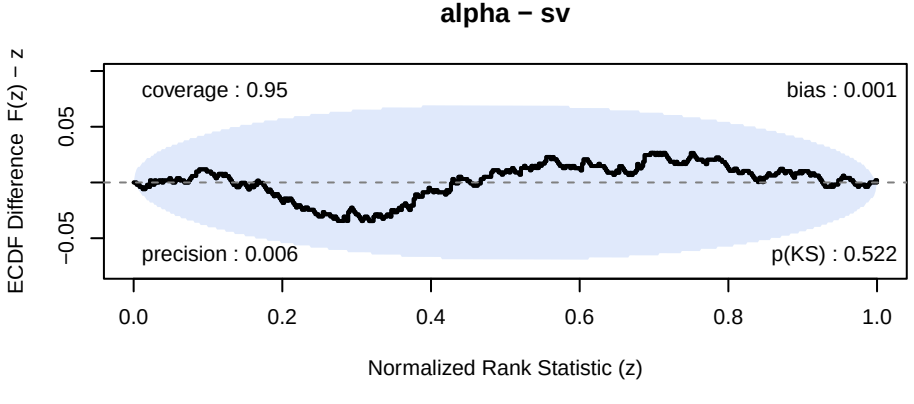
worst |std bias| 0.08 (t0); coverage 0.932−0.950; min KS p 0.231

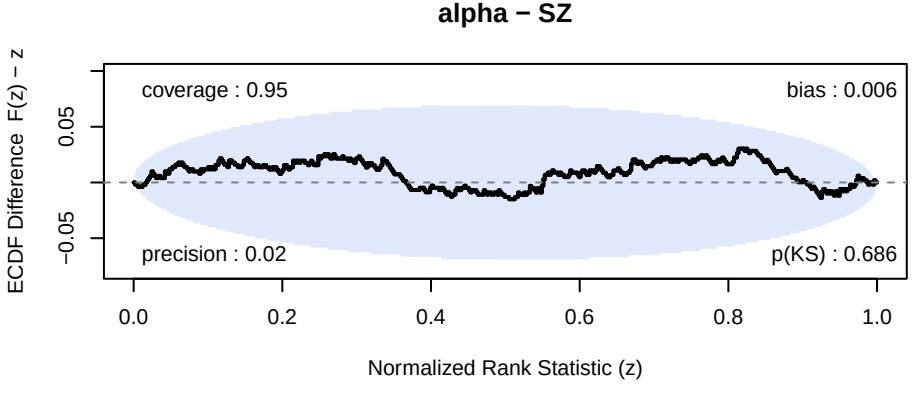

alpha − v
coverage : 0.94
bias : −0.002
precision : 0.119
p(KS) : 0.247
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − a
coverage : 0.93
bias : −0.001
precision : 0.147
p(KS) : 0.596
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.95
bias : 0
precision : 0.101
p(KS) : 0.231
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.93
bias : −0.001
precision : 0.044
p(KS) : 0.491
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − sv
coverage : 0.95
bias : 0.001
precision : 0.006
p(KS) : 0.522
ECDF Difference F(z) − z
Normalized Rank Statistic (z)

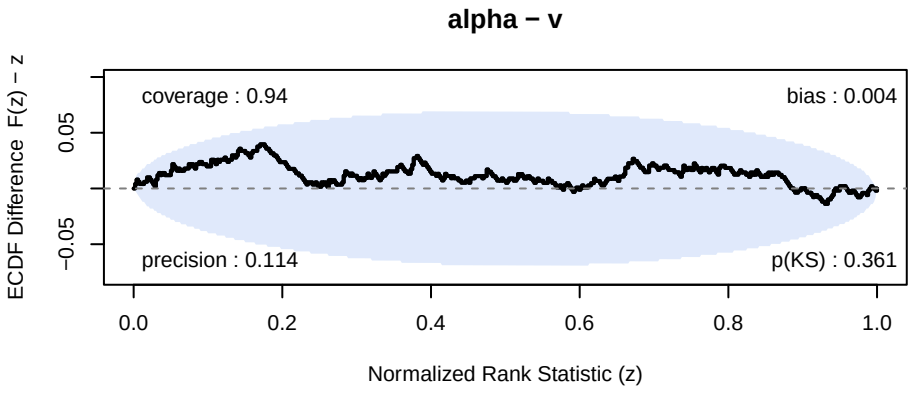

alpha − SZ
coverage : 0.95
bias : 0.006
precision : 0.02
p(KS) : 0.686
ECDF Difference F(z) − z
Normalized Rank Statistic (z)

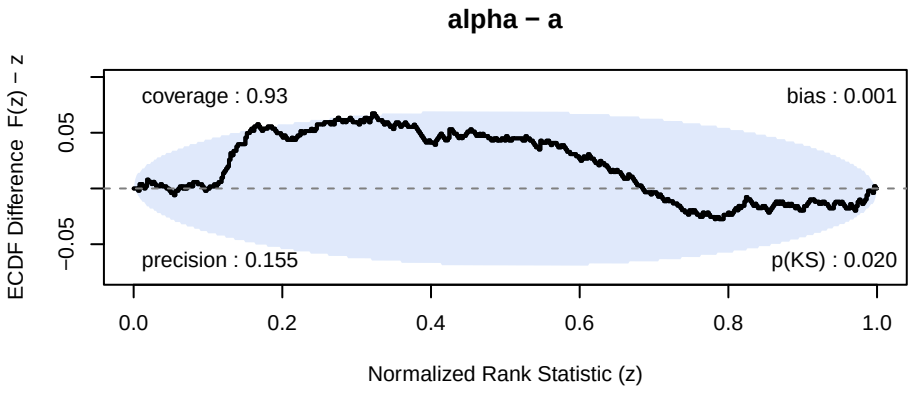
Case 5: st0 free (sv = SZ = 0)

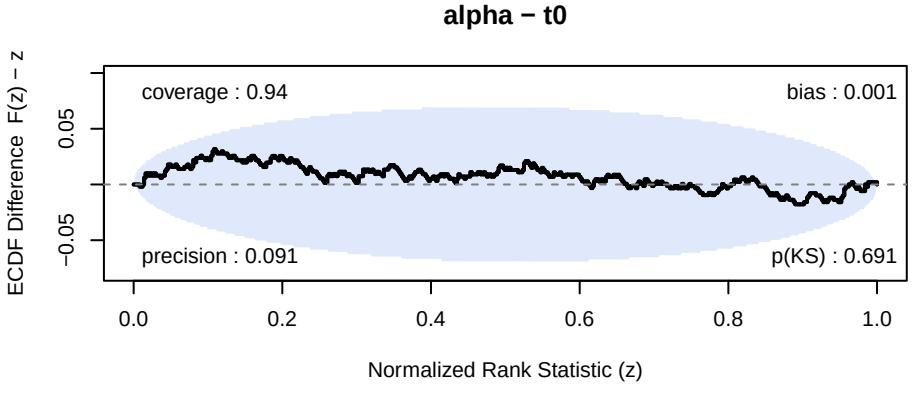
regime st0only

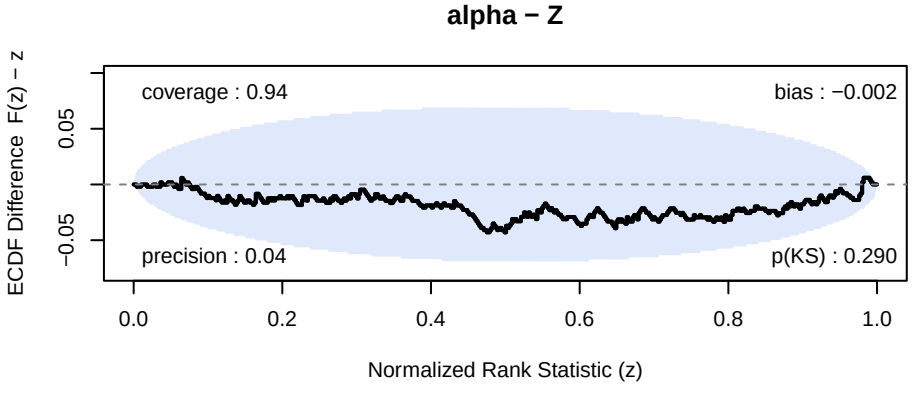
outside envelope: a

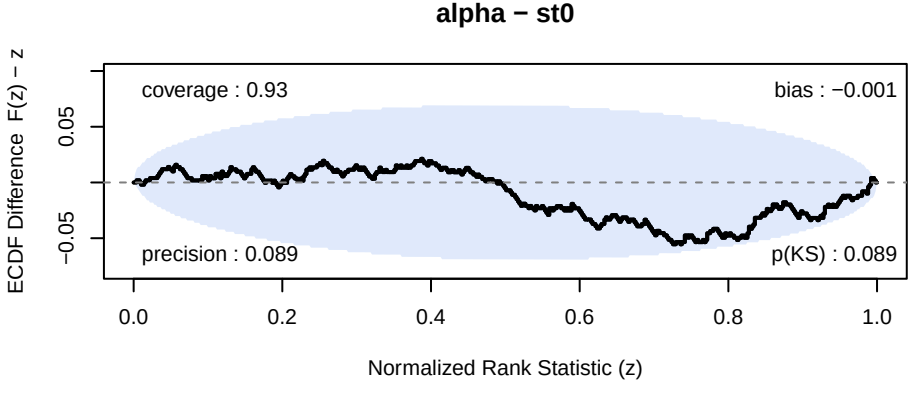
worst |std bias| 0.06 (Z); coverage 0.932−0.942; min KS p 0.020


alpha − v
coverage : 0.94
bias : 0.004
precision : 0.114
p(KS) : 0.361
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − a
coverage : 0.93
bias : 0.001
precision : 0.155
p(KS) : 0.020
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.94
bias : 0.001
precision : 0.091
p(KS) : 0.691
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.94
bias : −0.002
precision : 0.04
p(KS) : 0.290
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

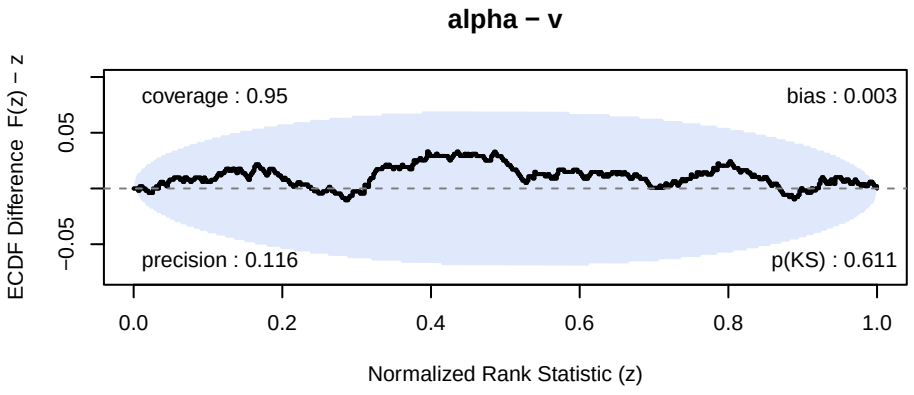

alpha − st0
coverage : 0.93
bias : −0.001
precision : 0.089
p(KS) : 0.089
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

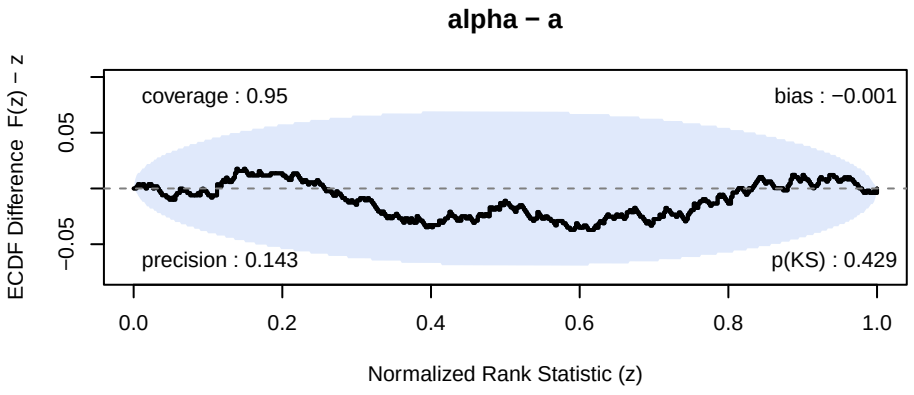
Case 7: SZ free (sv = st0 = 0)

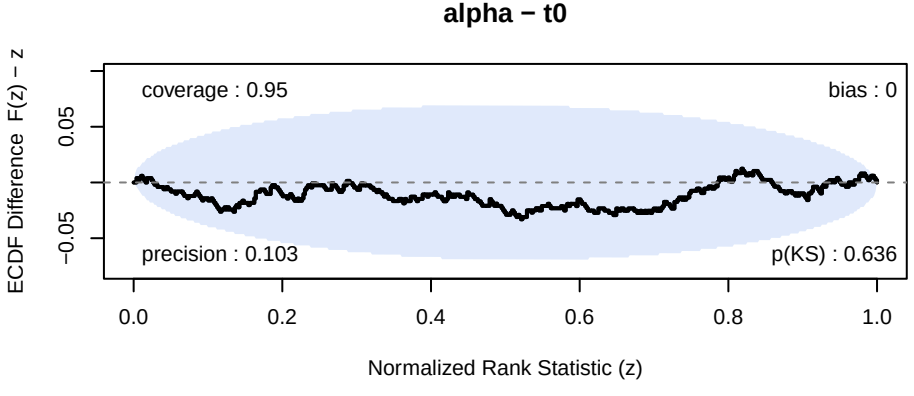
regime SZonly

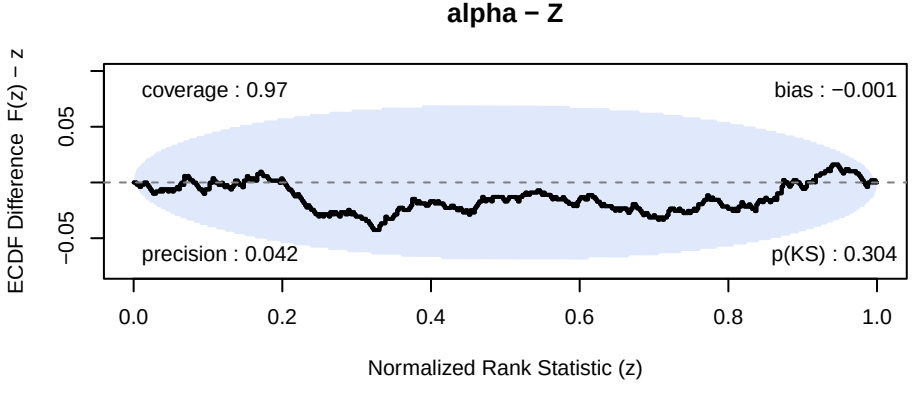
outside envelope: none

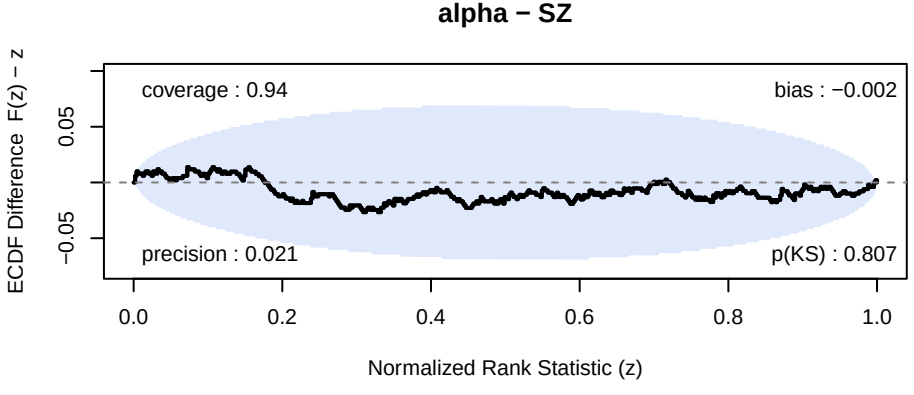
worst |std bias| 0.04 (Z); coverage 0.938−0.966; min KS p 0.304


alpha − v
coverage : 0.95
bias : 0.003
precision : 0.116
p(KS) : 0.611
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − a
coverage : 0.95
bias : −0.001
precision : 0.143
p(KS) : 0.429
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.95
bias : 0
precision : 0.103
p(KS) : 0.636
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.97
bias : −0.001
precision : 0.042
p(KS) : 0.304
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

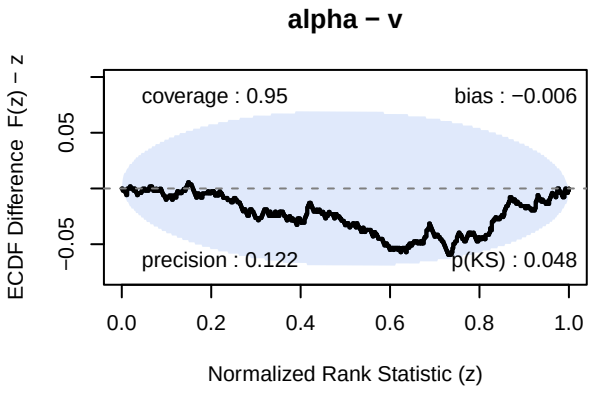

alpha − SZ
coverage : 0.94
bias : −0.002
precision : 0.021
p(KS) : 0.807
ECDF Difference F(z) − z
0.05
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Rank Statistic (z)

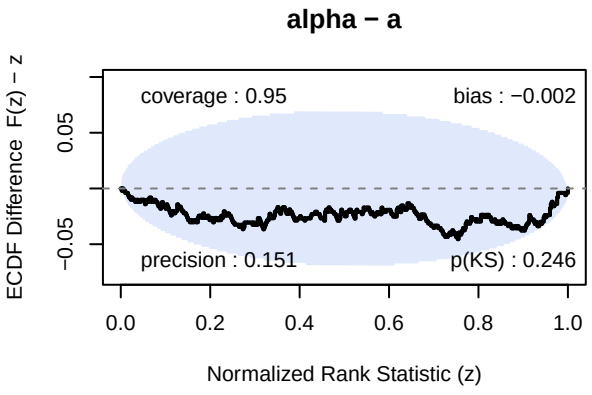
Case 8: all free

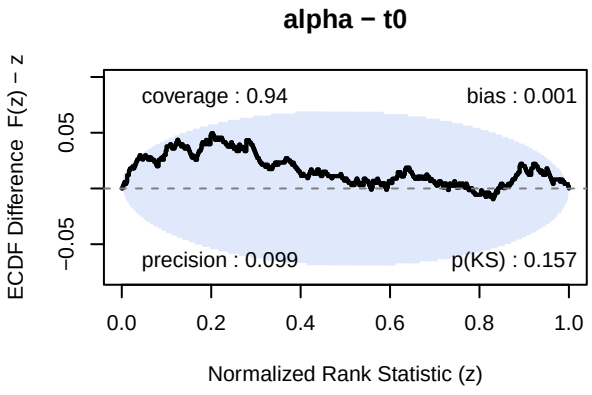
regime full

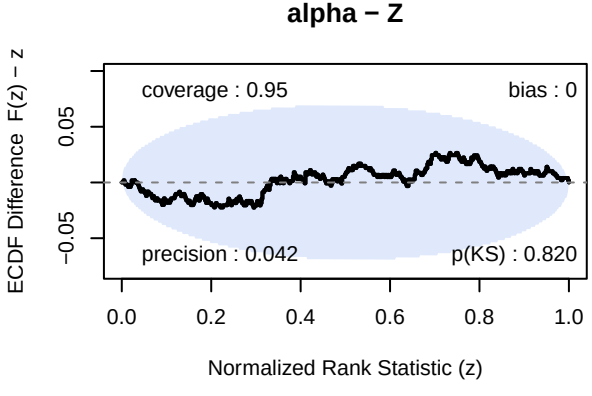
outside envelope: none

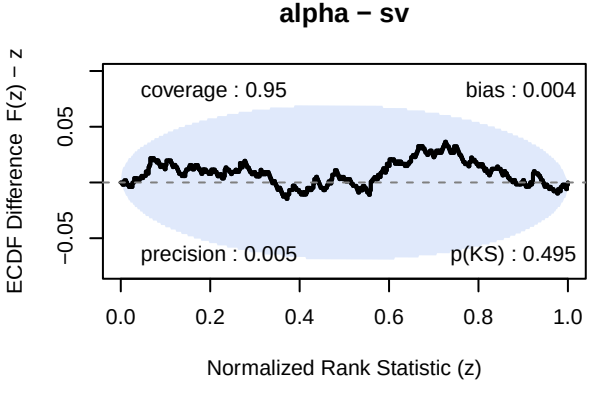
worst |std bias| 0.13 (t0); coverage 0.932−0.952; min KS p 0.048

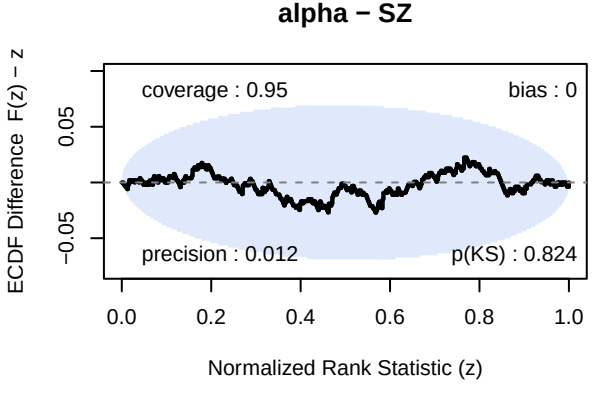

alpha − v
coverage : 0.95
bias : −0.006
precision : 0.122
p(KS) : 0.048
ECDF Difference F(z) − z
Normalized Rank Statistic (z)

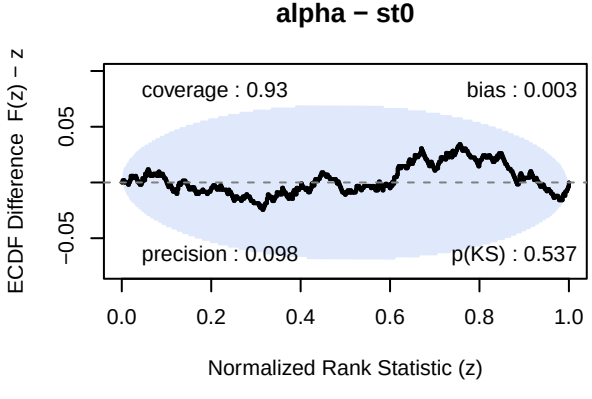

alpha − a
coverage : 0.95
bias : −0.002
precision : 0.151
p(KS) : 0.246
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − t0
coverage : 0.94
bias : 0.001
precision : 0.099
p(KS) : 0.157
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − Z
coverage : 0.95
bias : 0
precision : 0.042
p(KS) : 0.820
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − sv
coverage : 0.95
bias : 0.004
precision : 0.005
p(KS) : 0.495
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − SZ
coverage : 0.95
bias : 0
precision : 0.012
p(KS) : 0.824
ECDF Difference F(z) − z
Normalized Rank Statistic (z)


alpha − st0
coverage : 0.93
bias : 0.003
precision : 0.098
p(KS) : 0.537
ECDF Difference F(z) − z
Normalized Rank Statistic (z)